

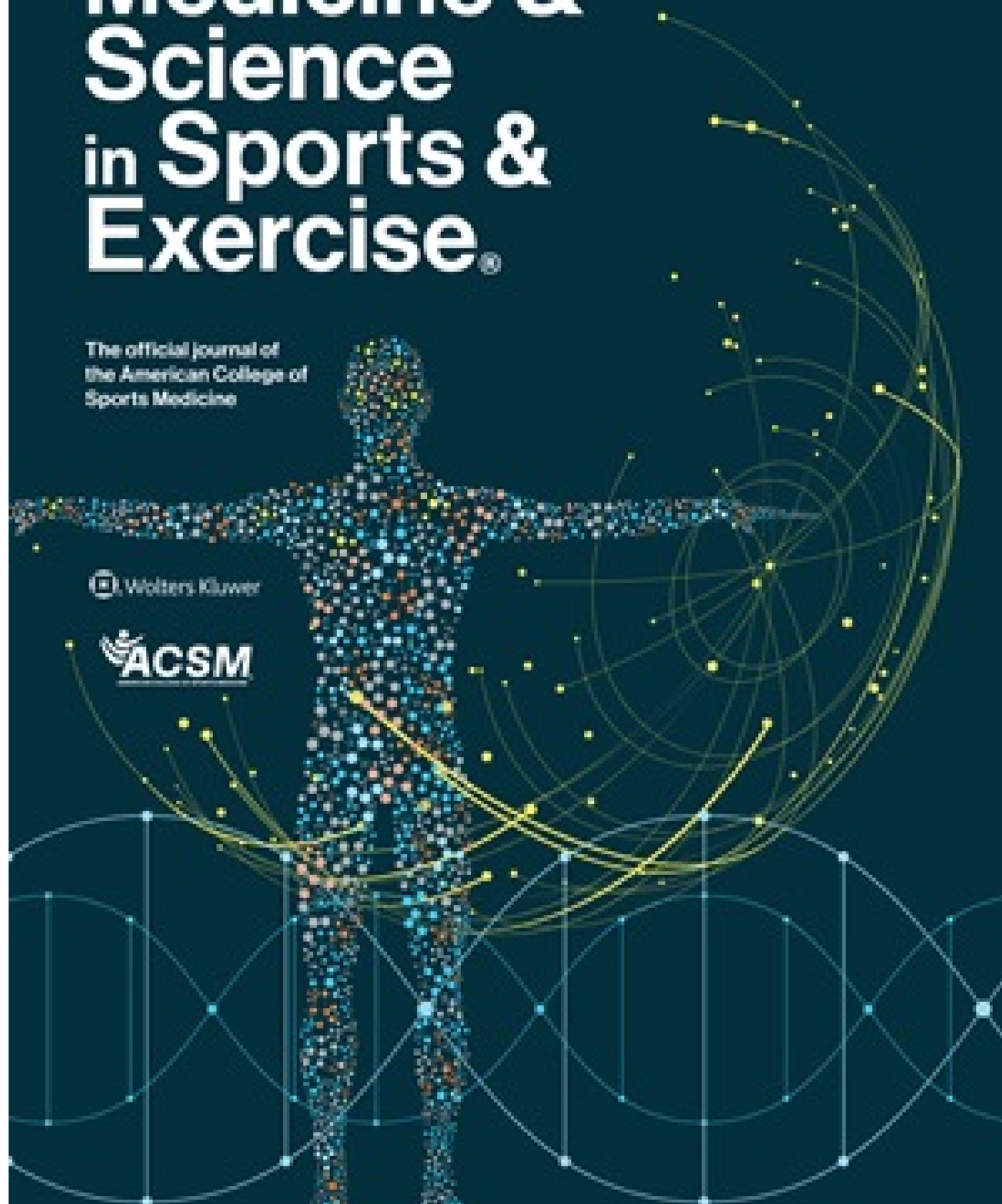
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Evidence for an inadequate hyperventilation inducing arterial hypoxemia at submaximal exercise in all highly trained endurance athletes

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Abstract

DURAND, F., P. MUCCI, and C. PRÉFAUT. Evidence for an inadequate hyperventilation inducing arterial hypoxemia at submaximal exercise in all highly trained endurance athletes. *Med. Sci. Sports Exerc.*, Vol. 32, No. 5, pp. 926–932, 2000.

The majority of highly trained endurance athletes with a maximal oxygen uptake greater than $60 \text{ mL} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$ develop exercise-induced hypoxemia (EIH). Yet some of them apparently do not. The pathophysiology of EIH seems to be multifactorial, and one explanatory hypothesis is a relative hypoventilation. Nevertheless, conflicting results have been reported concerning its contribution to EIH. The aim of this study was to compare the cardiorespiratory responses to maximal exercise of highly trained endurance athletes demonstrating the same aerobic capacity without EIH (N athletes) and with EIH (H athletes).

Ten N athletes and twelve H athletes performed an incremental exercise test. Measurements of arterial blood gases and cardiorespiratory parameters were performed at rest and during exercise.

All athletes presented a significant decrease in PaO_2 ($P < 0.05$) from rest up to 80% $\dot{\text{V}}\text{O}_{2\text{max}}$ associated with an increase in PaCO_2 , both findings consistent with a relative hypoventilation. Then the H athletes, who had a greater training volume per week and a higher second ventilatory threshold than the N athletes (respectively, 17 ± 1.1 vs $13.1 \pm 0.7 \text{ h} \cdot \text{wk}^{-1}$; 91.8 ± 1.7 vs $86.1 \pm 1.8\%$ $\dot{\text{V}}\text{O}_{2\text{max}}$), presented a continuous PaO_2 decrease up to $\dot{\text{V}}\text{O}_{2\text{max}}$. This was associated with a widening $(\text{A}_i\text{-a})\text{DO}_2$.

This study showed that a relative hypoventilation, probably induced by a high level of endurance training, induced hypoxemia in all athletes. However, a nonventilatory mechanism, perhaps related to the volume of training, seemed to affect gas exchanges beyond the second ventilatory threshold in the H athletes, thereby enhancing EIH.

It has been generally accepted that performance of intense dynamic exercise in highly trained endurance athletes is associated with reduced arterial pressure in O_2 (PaO_2) and an increased ideal alveolar-arterial PO_2 difference $(\text{A}_i\text{-a})\text{DO}_2$ ([10,12,27,32](#)). This exercise-induced hypoxemia (EIH) has been well documented in young athletes, but the pathophysiology is still unclear. Three potential mechanisms are thought to contribute to EIH: 1) a lack of compensatory hyperpnea, 2) a $\text{V}_\text{A}/\text{Q}$ mismatching, and/or 3) a diffusion limitation. Studies have suggested that the inadequate hyperventilation acts essentially for submaximal exercise ([28](#)) and that the $\text{V}_\text{A}/\text{Q}$ mismatching and the diffusion limitation appear for near-maximal levels of exercise ([16,28,32](#)).

One intriguing finding has not yet been studied, however. The majority of athletes who have a maximal oxygen uptake ($\dot{\text{V}}\text{O}_{2\text{max}}$) greater than $60 \text{ mL} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$ and who have been endurance training for several years develop this EIH. Yet some identically trained athletes do not exhibit it, and in some cases these athletes have shown a frank decrease in PaO_2 at submaximal workload and then an increase at maximal exercise during incremental testing (10; C. Préfaut and F. Durand, personal observations). It is possible, therefore, that near maximal exercise some physiological process either further impairs or improves arterial hypoxemia, resulting in no exercise-induced hypoxemia in some athletes. Few reports have compared endurance-trained athletes with the same aerobic capacity with and without the development of EIH, and studies that have examined the ventilatory hypothesis showed conflicting results about its contribution to EIH. Some studies indicated that an inadequate hyperventilation entirely explains EIH ([10,22](#)).

Another showed that this hypothesis holds only at submaximal exercise and thus explains only a part of EIH ([26](#)).

One hypothesis that would explain the divergent data in the literature is that a lack of compensatory hyperpnea exists in all endurance athletes but only at submaximal exercise. To better understand the mechanisms underlying exercise-induced hypoxemia, this study compared the cardiorespiratory responses with an incremental exercise of two groups of highly trained endurance athletes with the same aerobic capacity: one demonstrating exercise-induced hypoxemia, the other one not.

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METHODS

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Subjects

Twenty-two endurance-trained athletes participated in the study. No subject reported respiratory or cardiac disease or was known to be suffering from any chronic disease. None was on any regular medication. Before admittance to the study, all subjects were evaluated for cardiorespiratory health. All performed a preliminary maximal exercise test on a cycle ergometer to determine those who exhibited EIH. The criteria for selection of athletes were age < 30 yr and $\dot{V}O_{2\max} > 60 \text{ mL} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$. The study was approved by the institutional ethics committee and all subjects gave written consent to participate in the study after the protocol was fully explained to them.

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Athletes with exercise-induced hypoxemia (H).

Twelve male endurance-trained athletes, aged 24.2 yr, were studied. Six athletes were cyclists and six were triathletes. All athletes had been training regularly for 5.9 yr. EIH was defined as a fall in PaO_2 during exercise of more than 10 mm Hg below resting values that persists throughout at least the last three load levels of maximal exercise.

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Athletes without exercise-induced hypoxemia (N).

Ten male endurance-trained athletes, aged 23.5 yr, were studied. Five of the ten athletes were cyclists, four were triathletes, and one was a long distance runner. All had been training regularly for 5.1 yr. They did not show a hypoxemia maintained during at least three load levels of maximal exercise.

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Exercise Testing

The maximal exercise test was performed in the laboratory on a cycle ergometer (Monark 841, Varberg, Sweden). The test began with a warm-up of 3 min at 60 W. Pedaling speed remained constant (60 rpm) throughout the test, and the load was increased by 30 W every minute until exhaustion. The load increase was reduced at the last stages to allow the maximal power ($W_{\max}$) and the maximal oxygen uptake ($\dot{V}O_{2\max}$) to be assessed more accurately. Oxygen uptake was considered maximal if three of the following criteria were met: leveling off of $\dot{V}O_2$ despite increasing load, RER greater than 1.10, attainment of age-predicted maximal heart rate (HR) ($210 - (0.65 \text{ age})$), and inability of the subject to maintain the pedaling frequency despite maximum effort and verbal encouragement. A three-lead ECG (DII, V2, V5) (Quinton Q 3000, Seattle, WA) was recorded continuously during the exercise test.

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Cardiorespiratory Parameters

The ventilatory variables and the gas exchanges were measured continuously using a breath-by-breath automated exercise metabolic system (CPX Medical Graphics, St. Paul, MN). The subjects breathed through a low-resistance breathing valve (2700, Hans-Rudolph, Kansas City, KS). Before each test, the volume was calibrated by five inspiratory and expiratory strokes with a 3-L pump, and the gas analyzers were calibrated with two mixtures of gases of known O_2 and CO_2 concentration. Expired gases were analyzed for O_2 with a zirconia solid electrolyte O_2 analyzer and for CO_2 with an infrared analyzer. O_2 uptake ($\dot{V}O_2$), CO_2 output ($\dot{V}CO_2$), ventilation ($\dot{V}_E$) and its components tidal volume (V_T), breathing frequency (f), and respiratory exchange ratio (RER) were averaged during the last 20 s of each load over an integral number of breaths. They allowed us to calculate $\dot{V}_E/\dot{V}O_2$ and $V_E/\dot{V}CO_2$. The ratio V_D/V_T , the ventilatory reserves (VR) and the maximal oxygen pulse (O_2 pulse) were also calculated.

The two ventilatory thresholds (VT_1 and VT_2) were determined visually by two investigators using a double-blind procedure. Both were defined from the time course curves of $\dot{V}_E$ and the ventilatory equivalents of O_2 and CO_2 . The first ventilatory threshold, VT_1 , was determined by a nonlinear increase in ventilation ($\dot{V}_E$) and in the ventilatory equivalents of O_2 ($\dot{V}_E/\dot{V}O_2$), without increase in the ventilatory equivalents of CO_2 ($\dot{V}_E/\dot{V}CO_2$) ([34,35](#)). The second threshold, VT_2 , was identified by the second nonlinear increase in $\dot{V}_E$ and $\dot{V}_E/\dot{V}O_2$ accompanied by a concomitant nonlinear increase in $\dot{V}_E/\dot{V}CO_2$ ([1,30,31,33,36](#)). Thresholds were expressed in terms of $\dot{V}O_2$.

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Blood Gases

During a preliminary visit to the laboratory, blood gases to evaluate EIH were collected at the ear of the subject and were analyzed immediately for arterial PO_2 (PaO_2), PCO_2 ($PaCO_2$) and pH using the IL Meter 1306 (Milan, Italy).

On the day of the experiment, blood samples were drawn from the brachial artery of the nondominant arm with a catheter (Seldicath ϕ 1 mm, Plastimed, Paris, France) during the last 20 s of each load level of the protocol. Samples were analyzed as described above. To calculate the ideal partial pressure of alveolar O_2 ($PAiO_2$), we used the standard equation ([2](#)) where partial pressure of alveolar CO_2 has been estimated from $PaCO_2$ ([37](#)). We then calculated the ideal alveolo-arterial difference in O_2 ($(A_i-a)DO_2$).

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Protocol

During the preliminary visit to the laboratory, subjects also became acquainted with the personnel and equipment. On this first visit, clinical interviews and examinations were conducted: height, weight, blood pressure, lung spirometry, and resting ECG tracing were recorded. During the preliminary exercise test, we measured $\dot{V}O_{2max}$ and evaluated EIH. On the day of the experiment, the subjects again performed the exercise test described above. They were monitored for ventilatory and cardiac parameters throughout exercise testing and 5 min of the recovery. During testing, blood samples were drawn at rest and at 40, 50, 60, 70, 80, 90, and 100% $\dot{V}O_{2max}$.

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Statistics

The results are expressed as means $\pm$ SEM. After verification of a normal distribution, the physical, spirometric, and maximal data were compared between the N and H groups using a Student's *t*-test. Other data were compared using a two-way analysis of variance (ANOVA) with repeated measurement for between- and within-group differences. When the interaction of the two-way with repeated measurements was not significant, we used a one-way ANOVA with repeated measurements for within-group differences. Statistical significance was defined at the $P < 0.05$ level

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RESULTS

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Resting Characteristics

[Table 1](#) shows the group values for anthropometric and spirometric characteristics. There was no significant between-group difference.

TABLE 1. Physical and spirometric characteristics of athletes with and without EIH. Values are mean $\pm$ SEM.

	Age (yr)	Height (cm)	Body Mass (kg)	FEV ₁ (l)
Athletes with EIH	24.2 $\pm$ 1.3	182.2 $\pm$ 1.7	72.7 $\pm$ 1.5	4.7 $\pm$ 0.2
Athletes without EIH	23.5 $\pm$ 1	179.4 $\pm$ 2	70.8 $\pm$ 2.7	4.8 $\pm$ 0.1

Physical and spirometric characteristics of athletes with and without EIH. Values are mean $\pm$ SEM.

As described above, all subjects met the criteria for high-level endurance training status. All had been competing in their sport for the same number of years, but the H athletes had a greater overall training volume than the N athletes. Both groups had restarted their training for the competitive season approximately 4 months before the experimental test ([Table 2](#)).

TABLE 2. Characteristics of training in athletes with and without EIH. Values are mean $\pm$ SEM.

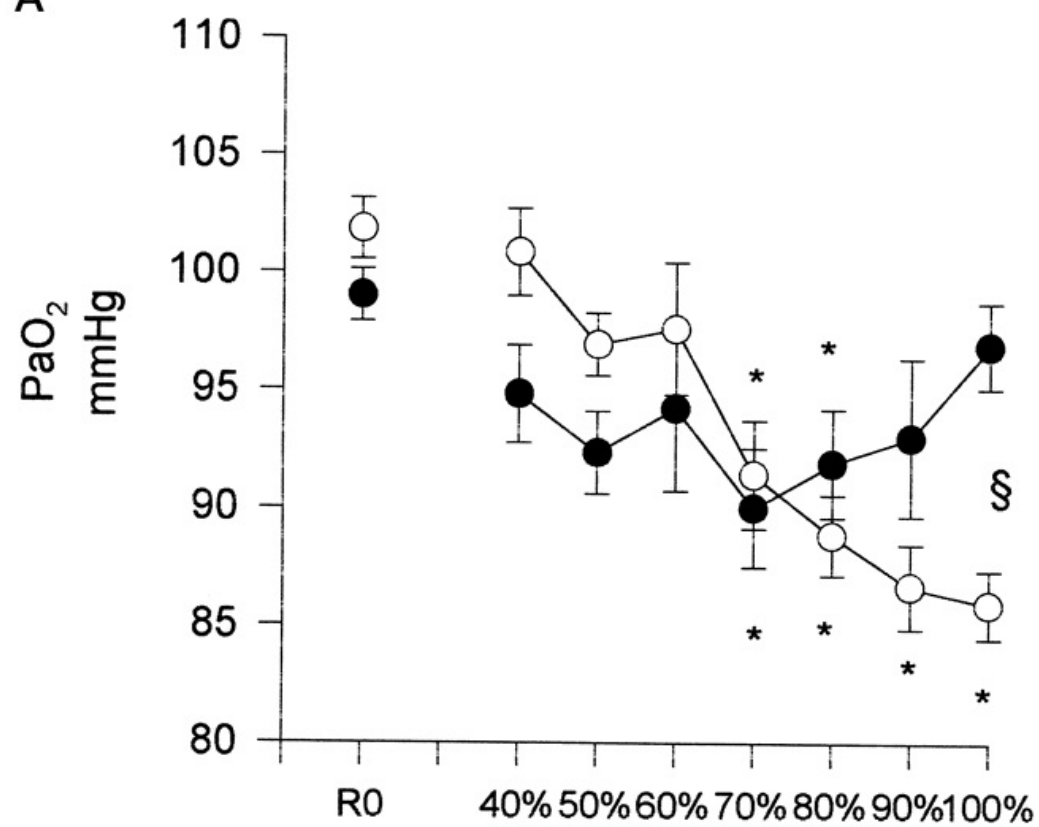
	Training (h·wk ⁻¹)	Time in competition (yr)	Training Since Re-opening of Season (months)
Athletes with EIH	17 $\pm$ 1.1*	5.9 $\pm$ 1.3	4.8 $\pm$ 1.4
Athletes without EIH	13.1 $\pm$ 0.7	5.1 $\pm$ 0.9	3.7 $\pm$ 1.7

* Statistical difference; $P < 0.05$.

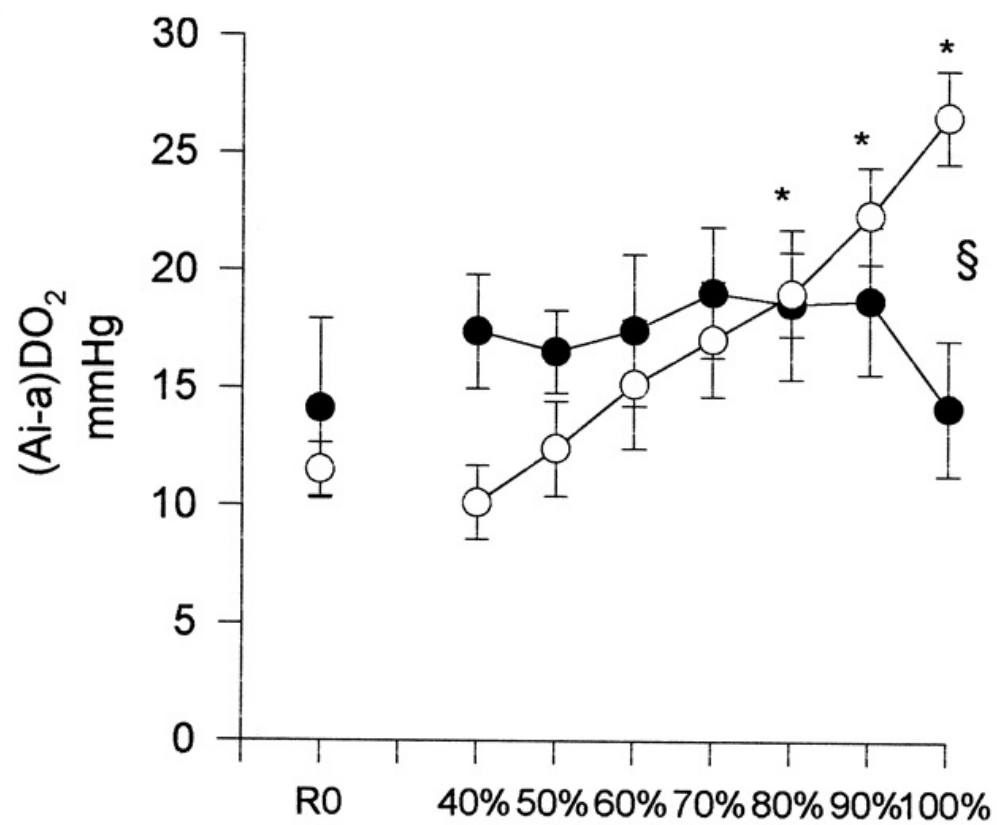
Characteristics of training in athletes with and without EIH. Values are mean $\pm$ SEM.

At rest, there was no difference in PaO₂, (A_i-a)DO₂, PaCO₂, PAiO₂ ([Figs. 1 and 2](#)) or ventilatory parameters $\dot{V}_E$, $\dot{V}_E/\dot{V}O_2$, $\dot{V}_E/\dot{V}CO_2$, f , V_T and V_D/V_T between the two groups of athletes ([Table 3](#)).

A



B



Time course of PaO₂ (A) and (A_i-a)DO₂ (B) during rest and exercise with (○) and without (•) EIH. * statistical difference observed with resting values within groups; P < 0.05. § statistical difference observed between groups; P < 0.05.

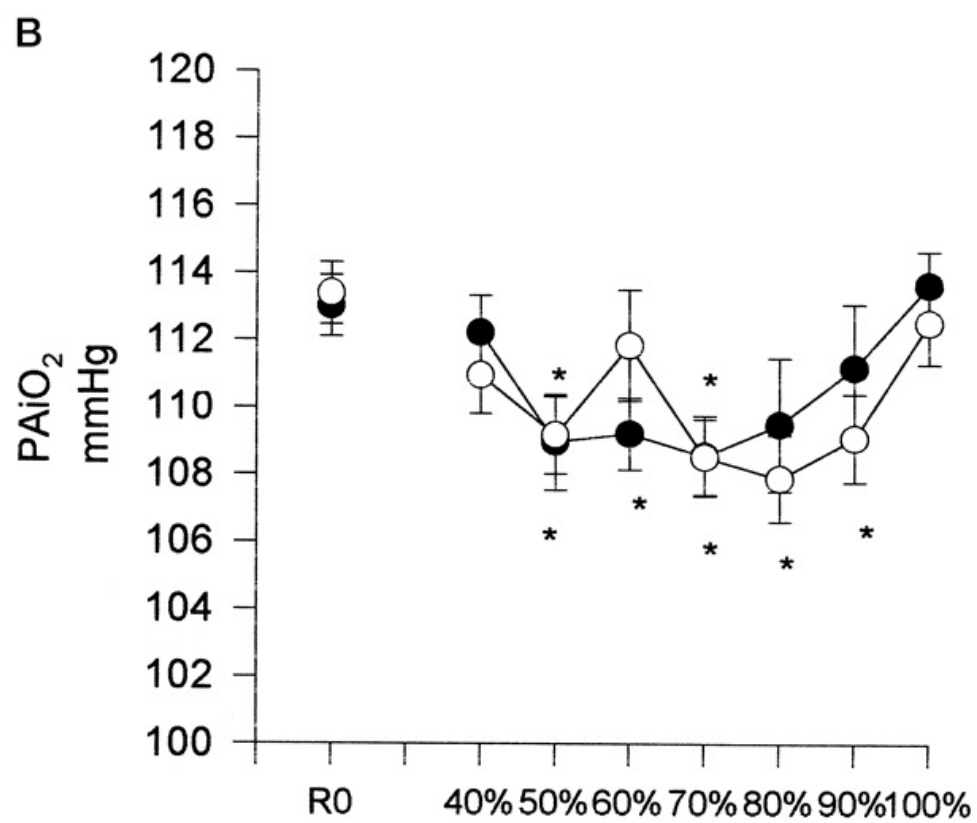
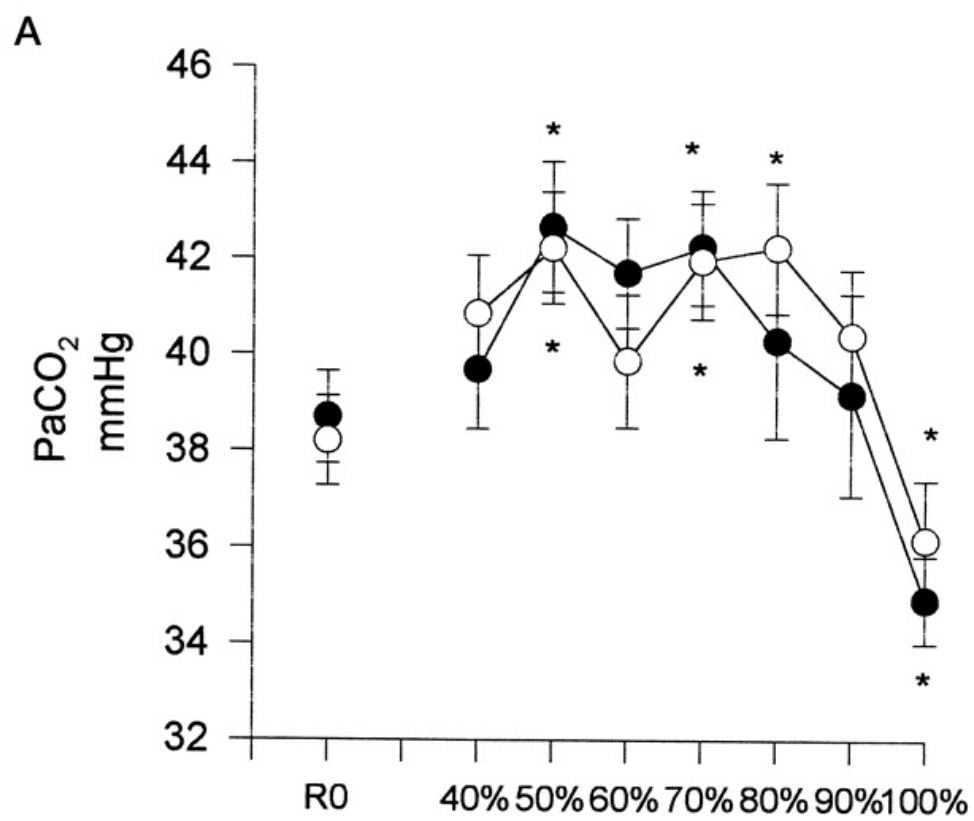


TABLE 3. Ventilatory parameters during exercise in athletes with and without EIH. f, breathing rate; $\dot{V}_E$, minute ventilation; $\dot{V}_E/\dot{V}O_2$, respiratory equivalent in O₂; $\dot{V}_E/\dot{V}CO_2$, respiratory equivalent in CO₂. Values are mean $\pm$ SEM.

Athletes	Rest	Exercise (% $\dot{V}O_{2max}$)						
		40	50	60	70	80	90	100
$\dot{V}_E$ (L·min ⁻¹)								
EIH	13.3 $\pm$ 0.7	43.2 $\pm$ 2.2	52.6 $\pm$ 1.8†	65.8 $\pm$ 3.7†	82.3 $\pm$ 4.3†	97.6 $\pm$ 5.5†	122.4 $\pm$ 6†	171 $\pm$ 7.1†*
No EIH	12.8 $\pm$ 0.9	40 $\pm$ 2.8	47.6 $\pm$ 3.8†	63 $\pm$ 4.1†	70.5 $\pm$ 5.7†	94.5 $\pm$ 6.1†	116.6 $\pm$ 9.9†	171.1 $\pm$ 4†*
$\dot{V}_E/\dot{V}O_2$								
EIH	28.3 $\pm$ 1.2	18.4 $\pm$ 0.6	18.6 $\pm$ 0.6	18.9 $\pm$ 0.8	20.8 $\pm$ 1.4†	21.4 $\pm$ 1.3†	23.8 $\pm$ 1.2†	30.3 $\pm$ 1.2†*
No EIH	27 $\pm$ 1.5	18.1 $\pm$ 0.6	18.2 $\pm$ 0.7	18.7 $\pm$ 0.7	19.1 $\pm$ 0.7†	21.2 $\pm$ 0.8†	23.4 $\pm$ 1.2†	31.6 $\pm$ 0.8†*
$\dot{V}_E/\dot{V}CO_2$								
EIH	33.6 $\pm$ 1.3	22.3 $\pm$ 0.6	21.6 $\pm$ 0.6†	21.2 $\pm$ 0.6	21.5 $\pm$ 0.7	21.3 $\pm$ 0.7	22.1 $\pm$ 0.7†	25.1 $\pm$ 0.7†*
No EIH	33.2 $\pm$ 1.4	22 $\pm$ 0.4	21.4 $\pm$ 0.4†	20.6 $\pm$ 0.3	20.5 $\pm$ 0.3	18.8 $\pm$ 2.1	21.5 $\pm$ 0.7	25.5 $\pm$ 0.6†*
f (breaths·min ⁻¹)								
EIH	14.9 $\pm$ 0.8	22.2 $\pm$ 1.4	23.3 $\pm$ 1.3†	25.8 $\pm$ 1.4†	30 $\pm$ 1.7†	31.9 $\pm$ 1.5†	37 $\pm$ 1.5†	52.6 $\pm$ 2.2†*
No EIH	14.7 $\pm$ 0.6	20.2 $\pm$ 1	22.5 $\pm$ 1.4	22.2 $\pm$ 2.8	26.4 $\pm$ 1.3†	31.8 $\pm$ 1.5†	36.6 $\pm$ 2†	53.4 $\pm$ 1.8†*
V_T (L)								
EIH	0.92 $\pm$ 0.06	1.96 $\pm$ 0.11†	2.29 $\pm$ 0.11†	2.53 $\pm$ 0.14†	2.74 $\pm$ 0.1	3.01 $\pm$ 0.11†	3.23 $\pm$ 0.1	3.24 $\pm$ 0.1*
No EIH	0.89 $\pm$ 0.06	2 $\pm$ 0.16†	2.21 $\pm$ 0.15†	2.63 $\pm$ 0.18†	2.81 $\pm$ 0.21	2.99 $\pm$ 0.21	3.2 $\pm$ 0.19	3.19 $\pm$ 0.12*
V_D/V_T (%)								
EIH	0.42 $\pm$ 0.02	0.2 $\pm$ 0.01†	0.19 $\pm$ 0.02	0.14 $\pm$ 0.02	0.18 $\pm$ 0.01	0.18 $\pm$ 0.01	0.18 $\pm$ 0.01	0.19 $\pm$ 0.02*
No EIH	0.42 $\pm$ 0.03	0.17 $\pm$ 0.02†	0.2 $\pm$ 0.03	0.15 $\pm$ 0.02	0.15 $\pm$ 0.02	0.15 $\pm$ 0.02	0.17 $\pm$ 0.03	0.18 $\pm$ 0.02*

* Statistical change within group during exercise test; P < 0.001. † Statistical difference with precedent level of exercise within group; P < 0.05.

Ventilatory parameters during exercise in athletes with and without EIH. f, breathing rate; $\dot{V}_E$, minute ventilation; $\dot{V}_E/\dot{V}O_2$, respiratory equivalent in O₂; $\dot{V}_E/\dot{V}CO_2$, respiratory equivalent in CO₂. Values are mean $\pm$ SEM.

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Blood Gases During Exercise

The H group showed a significant decrease in PaO₂ (P < 0.05) during exercise testing from 101.8 $\pm$ 1.3 mm Hg measured at rest to 85.9 $\pm$ 1.4 mm Hg at maximal exercise ([Fig. 1A](#)). Each H athlete showed a minimal decrease in PaO₂ of 10 mm Hg, and the mean value of this drop was 16.7 $\pm$ 1.3 mm Hg. The decrease in PaO₂ was significant from 70% $\dot{V}O_{2max}$. Within this group, (Ai-a)DO₂ increased significantly (P < 0.05) from 11.5 $\pm$ 1.1 mm Hg measured at rest to 26.6 $\pm$ 2 mm Hg at maximal exercise ([Fig. 1B](#)). This increase was significant from 80% $\dot{V}O_{2max}$. The N group showed no statistical decrease in PaO₂ between rest and maximal exercise ([Fig. 1A](#)). We did, however, observe a significant marked PaO₂ decrease from rest (99 $\pm$ 1.1 mm Hg) to 70% $\dot{V}O_{2max}$ (90 $\pm$ 2.5 mm Hg) (P < 0.05). Beyond 70% $\dot{V}O_{2max}$, we observed a tendency for PaO₂ to increase. The N group showed no significant increase in (Ai-a)DO₂, only a tendency ([Fig. 1B](#)). A between-group difference for PaO₂ and (Ai-a)DO₂ was observed at maximal exercise (P < 0.05).

During exercise we observed significant changes in PaCO₂ and PAiO₂ in both groups ([Figs. 2A and B](#)). Within each group a significant increase in PaCO₂ (>40 mm Hg) and a concomitant decrease in PAiO₂ (<110 mm Hg) occurred from 50% $\dot{V}O_{2max}$ (P < 0.05).

Then, a significant decrease in PaCO₂ (close to 35 mm Hg) and a significant increase in PAiO₂ (>112 mm Hg) occurred from 70% $\dot{V}O_{2max}$ for the N group, and later for the H group, until maximal exercise ($P < 0.05$). There were no between-group differences for these variables.

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Cardiorespiratory Responses to Incremental Exercise

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Global response.

$\dot{V}O_{2max}$ and W_{max} reached high values for both H and N athletes, and there was no between-group difference ([Table 4](#)). $\dot{V}_E/\dot{V}O_2$ and $\dot{V}_E/\dot{V}CO_2$ increased significantly during exercise in the two groups ($P < 0.001$) and no between-group difference was observed ([Table 3](#)).

TABLE 4. Maximal data, first (VT₁) and second (VT₂) ventilatory thresholds recorded during maximal incremental exercise testing in athletes with and without EIH. Values are mean ± SEM.

	$\dot{V}O_{2max}$ (ml·min ⁻¹ ·kg ⁻¹)	Maximal load (W)	Maximal HR (beats·min ⁻¹)	O2 pulse (mL·beat ⁻¹)	Ventilatory Reserves (%)	VT ₁ (%) $\dot{V}O_{2max}$	VT ₂ (%) $\dot{V}O_{2max}$
Athletes with EIH	64.6 ± 1.2	376.2 ± 8.9	183.2 ± 3.6	24.7 ± 0.6	4.53 ± 1.94	67.1 ± 1.4	91.8 ± 1.7*
Athletes without EIH	64.1 ± 1.2	361.5 ± 8.5	189.5 ± 3.6	23.8 ± 0.8	3.53 ± 1.78	66 ± 1.9	86.1 ± 1.8

* Statistical difference; $P < 0.05$.

Maximal data, first (VT1) and second (VT2) ventilatory thresholds recorded during maximal incremental exercise testing in athletes with and without EIH. Values are mean ± SEM.

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Ventilatory response.

$\dot{V}_E$, f , and V_T increased significantly ($P < 0.001$) during exercise testing in both groups and there was no difference between the H and N groups ([Table 3](#)). [Table 4](#) showed no difference in the two groups for ventilatory reserves, which were exhausted. There was no difference in VT₁, whereas VT₂ was significantly higher for H athletes than for N athletes ($P < 0.05$).

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Cardiac response.

At maximal exercise, HR and O₂ pulse presented no statistical difference between the two groups of athletes ([Table 4](#)). The ratio V_D/V_T decreased significantly during exercise testing in each group ($P < 0.001$) and no between-group difference was observed ([Table 3](#)).

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DISCUSSION

This study showed that all highly trained endurance athletes with the same $\dot{V}O_{2\max}$ exhibited a fall in PaO₂ during submaximal exercise. However, exercise-induced hypoxemia at the end of incremental exercise was present only in the group of athletes who showed an increasing (A_i-a)DO₂ and a later appearance of the second ventilatory threshold. Furthermore, the EIH athletes were the group with the greatest training volume per week.

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Methodology

The blood gas values in the present study are presented for a temperature of 37°C. We did not record body temperature to minimize invasive measurement in the same subject. Thus, arterial blood gas measurements were not corrected. Nevertheless, given the exercise regimen of our athletes and the absence of significant difference in $\dot{V}O_{2\max}$ and maximal workload in our two groups, we expected temperature to increase similarly in N and H athletes. To minimize the risk of overestimation induced by an increase in body temperature, we added 3 mm Hg to the 5 mm Hg drop in PaO₂ that is the usual definition of resting hypoxemia ([29](#)), i.e., a value of 8 mm Hg. The temperature increase was <0.6°C during a short exercise test ([9,38](#)), and this was confirmed by measurement of rectal temperature ([6](#)). Therefore, to account for the error caused by the lag between rectal temperature and blood temperature, i.e., 2 or 3 mm Hg ([14](#)), we defined exercise-induced hypoxemia as a drop of more than 10 mm Hg in PaO₂.

Ventilatory thresholds were determined using the respiratory gas exchange indices as described above. This gas exchange method has been validated by numerous studies for the first ventilatory threshold ([7,19](#)), which corresponds to blood lactate production. Recently it has been shown that the lactate accumulation threshold can be accurately estimated from measurements of respiratory gas exchanges ratio in young athletes ([1](#)).

Responses to Exercise Testing

The high values of $\dot{V}O_{2\max}$ and $W_{\max}$ reached by the two groups of athletes confirmed that these subjects represent a population of elite endurance athletes. From 70% $\dot{V}O_{2\max}$, the H group exhibited a marked and continuous decrease in PaO_2 during exercise testing. In this group, we noted four athletes with 10 to 15 mm Hg reductions in PaO_2 , five with 15 to 20 mm Hg reductions, two with 20 to 25 mm Hg reductions, and the remaining one with a reduction of more than 25 mm Hg. An associated increase in $(A_i-a)DO_2$ occurred in this population from 80% $\dot{V}O_{2\max}$, indicating that hypoxemia appears first and then is associated with a $(A_i-a)DO_2$ increase.

The main result of this study was that N athletes showed the same decrease in PaO_2 as H athletes up to 80% $\dot{V}O_{2\max}$, without a concomitant increase in $(A_i-a)DO_2$. The hypothesis that a ventilatory factor contributes substantially to EIH at submaximal exercise, which would be enhanced by a diffusion limitation at maximal levels of exercise, has already been suggested ([4,28](#)). When hypoxemia occurs during submaximal exercise, it has been consistently associated with a sluggish ventilatory response, with $PAiO_2$ values < 112 mm Hg and $PaCO_2$ values > 35 mm Hg ([10](#)). In both H and N groups, our data were consistent with this hypothesis of a lack of compensatory hyperpnea at submaximal exercise. This was further supported by low $\dot{V}E/\dot{V}O_2$ and $\dot{V}E/\dot{V}CO_2$ values measured at the same time, while f and $\dot{V}_E$ increased regularly. In contrast with previous results ([13](#)), the N athletes presented the same criteria for a lack of compensatory hyperpnea as the H athletes at submaximal exercise. An alveolar ventilation below the rate metabolically required to maintain normal arterial blood gases values has been demonstrated in endurance-trained athletes at submaximal exercise ([8](#)). Because untrained subjects do not exhibit this phenomenon, it is probably at least partially a result of 1) the reduction in lactacidemia and acidosis induced by high levels of endurance training ([11](#)) and/or 2) the decrease in carbohydrate oxidation at moderate exercise ([3](#)). A recent study indicated that slower ventilation with increasing exercise intensities appears to be more closely linked to the decreased reliance on carbohydrate oxidation than to the attenuated increase in blood lactate concentration ([20](#)). Another explanation is a different ventilatory responsiveness, and some authors have reported that trained athletes present a low hypercapnic ventilatory responsiveness at rest ([13,22](#)) and a high threshold of ventilatory response to CO_2 during moderate exercise ([23](#)). It is thus likely that both the N and H athletes, all of whom had a high endurance-training status, presented a lack of compensatory hyperpnea

as the major cause of the hypoxemia recorded from rest up to 80% $\dot{V}O_{2\max}$, i.e., for submaximal exercise. It is widely acknowledged, however, that other factors, such as V_A/Q inequality and diffusion limitation, occur at the same time ([12,16,32](#)). Indeed, the results of our study are in agreement with those of two others, both of which concluded that an inadequate hyperventilatory response could not explain EIH at maximal exercise ([15,26](#)).

Between 80 and 90% $\dot{V}O_{2\max}$, near the VT_2 of the athletes, the time course of PaO_2 changed in both groups. The N athletes presented a later VT_2 , therefore an earlier hyperventilation, which probably explains the PaO_2 increase; however, $PaCO_2$ did not decrease significantly. The H athletes, who had a higher VT_2 than the N athletes (91.8 vs 86.1% $\dot{V}O_{2\max}$), presented a later hyperventilation. PaO_2 continued to decrease and a concomitant increase in $(A_i-a)DO_2$ occurred. We did not measure respiratory mechanics during this heavy exercise. However, both groups presented maximal ventilation close to, but under the limits of, their maximal voluntary ventilation, i.e., near the maximum flow volume limitation. We may thus assume that mechanical constraint was unlikely to act on exercise hyperpnea in our athletes, which is in agreement with a study of Johnson et al. ([17](#)). A delayed hyperventilatory response and/or a gas exchange alteration may thus have contributed to the persisting hypoxemia at maximal exercise in this population.

At maximal exercise, a gas exchange alteration was signaled by a widening in $(A_i-a)DO_2$, which contributed to the increased hypoxemia in H athletes. Several studies have indicated that nonventilatory phenomena such as V_A/Q mismatching and diffusion limitation occur in EIH ([12,21,32](#)) for maximal levels of exercise. V_A/Q mismatching is known to explain only one-third of hypoxemia ([12](#)), and the mechanism of diffusion limitation remains unclear. The hypothesis that diffusion limitation is caused by an interstitial edema, with an accumulation of water in the interstitium and the peribronchovascular space, is now well supported, but the cause of this edema is under debate ([5,12,31](#)).

According to the delayed VT_2 the role of training should also be considered. The sport specialties were nearly the same in the two groups, and EIH has already been reported in various endurance specialists, such as cyclists ([24](#)), triathletes ([4](#)), and middle- and long-distance runners ([10](#)), all of whom were represented in our groups of athletes. It is thus highly unlikely that sport specialty had an effect on this study. The H athletes reported a greater training volume per week, which would explain the higher VT_2 and which strongly suggests that EIH is at least partly dependent on the quantity of training. Furthermore, the similar $\dot{V}O_{2\max}$ in the two groups, despite the fact that H trained more,

suggests that EIH in athletes is sufficient to exert a limited effect on $\dot{V}O_{2\max}$, as already shown by Powers (25). Our data showed that our athletes had restarted training approximately the same number of months before the opening of the competitive season. This indicates that a marked difference in training intensity was unlikely, as we know that training schedules are similar during the competitive season. Nevertheless, we cannot confirm that the training intensity was the same in the two groups of athletes because we did not record their daily schedule of training. This would be an interesting perspective for a future study but difficult to monitor. Indeed, a study from Miyachi's group (18) reported that 12 wk of maximal interval training performed by active but not endurance-trained subjects contributed to the development of EIH. No correlation, however, was found between the amplitude of arterial hypoxemia and VT_2 , suggesting that delayed VT_2 is not the main cause of EIH.

In summary, this study indicates that highly trained athletes who do not develop exercise-induced hypoxemia show the same marked arterial hypoxia at submaximal exercise as athletes with EIH. This suggests that the same phenomenon, that is, a lack of compensatory hyperpnea induced by endurance training, occurs in the first part of EIH. At maximal exercise, however, we can posit a compensatory mechanism in the non-EIH athletes, since the submaximal hypoxemia was reversed. In contrast, beyond the second ventilatory threshold athletes with EIH present a nonventilatory phenomenon that appears to be dependent on the volume of training.

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